MNTC 313 – Assignment 11

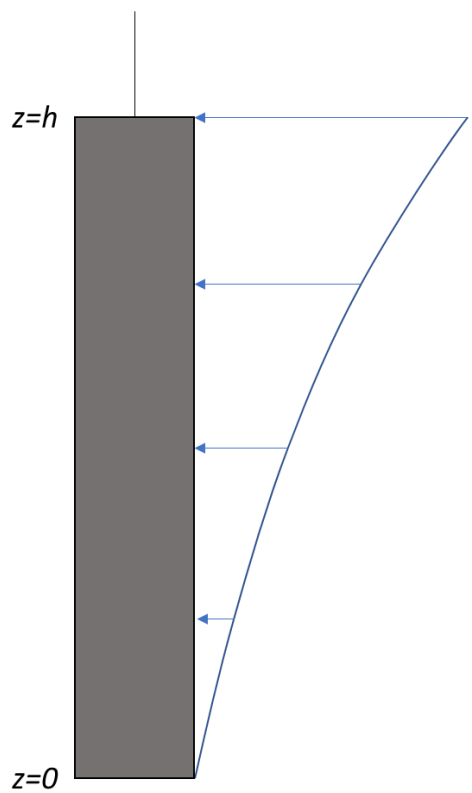
When designing large skyscrapers, wind is an important factor to take into consideration, as wind loading on the surfaces of the building can pose a significant threat to its stability. Wind speeds usually increase with altitude, and as modern skyscrapers are towering to heights of almost 1000 metres, engineers must ensure that the structure is able to withstand the wind force.

You are part of an engineering team that has been tasked with determining the wind force on a building that is 500 metres tall. At its peak, wind loading has been observed to follow the function below:

$$w(z) = 1.8^{0.045z} - 1$$

Where *w(z)* is the distributed load, in Newtons/metre, at a given height, *z*, in metres. Thus, the total force on a given portion of the building is represented by the integral of w(z) from point a to point b:

$$F = \int_a^b w(z)dz$$



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Using MATLAB, develop a program that does the following:

- Takes user input for the integration limits, a and b, then calculates and prints the true value of the integral. **HINT**: you have been given the function handle and user input in the starter code. You should use the **quadgk** function to evaluate the exact value.

- Uses the rectangular method to evaluate the integral numerically. You should conduct numerical integrations at N values of 10, 100, 1000, 10000 and 100000. **Please use the MATLAB method specified in the Week 11 lecture slides.**

- Calculates the percent error between the numerical value and the true value using the following equation:

$$\%\ Error = \left|\frac{Estimated\ Value - True\ Value}{True\ Value}\right| * 100$$

- Prints the N value used, the numerical integration value and the percent error

- **Complete this part after you have integrated with all five N values above.** Considering that the building is 500 metres tall, calculate the total **pressure** acting on the top 50 metres of the building assuming that the building is 50 metres wide. Use N = 10000 and recall that Pressure = Force/Area. SI Units for pressures are Pascals (Pa), which are N/m^2.

NOTE: A starter code file is provided on OnQ, which includes formatted user input statements and the function handle.

Reminder: Please **DO NOT** include any personal information such as name, student number or Net ID in your submission. All submissions to Gradescope will be run through plagiarism detection software.

Hint: other useful function is **linspace**

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An example of your output should look like:

```
Enter a:5

Enter b:20

True value: 6 N
N = 10, F = 6 N, Error % = 0.51
N = 100, F = 6 N, Error % = 0.05
N = 1000, F = 6 N, Error % = 0.00
N = 10000, F = 6 N, Error % = 0.00
N = 100000, F = 6 N, Error % = 0.00
Average pressure between 450 and 500 m: 6.1 kPa
```